General Chemistry II (CHEM1111)
Equation Cheat Sheet

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1 The Gaseous State

1.1 The Ideal Gas Law

$$PV = nRT \tag{1}$$

1.2 Dalton's Law of Partial Pressures

$$P_{\mathrm{A}} = X_{\mathrm{A}} P_{\mathrm{tot}} \tag{2}$$

2 Solutions

2.1 Composition of Solutions

2.1.1 Mole fraction

$$X_i = \frac{n_i}{\sum\limits_i n_i} \tag{3}$$

2.1.2 Molarity

$$M = \frac{n_{\mathrm{solute}}}{V_{\mathrm{solution}}} \tag{4}$$

2.1.3 Molality

$$m = \frac{n_{\mathrm{solute}}}{m_{\mathrm{solvent}}} \tag{5}$$

2.1.4 Dilution of solution

$$c_i V_i = c_f V_f \tag{6}$$

2.2 Titration

$$\frac{c_{\mathrm{A}} V_{\mathrm{A}}}{N_{\mathrm{A}}} = \frac{c_{\mathrm{B}} V_{\mathrm{B}}}{N_{\mathrm{B}}} \tag{7}$$

2.3 Colligative Properties

2.3.1 Raoult's Law

$$P_1 = X_1 P_1^{\circ} \tag{8}$$

2.3.2 Boiling-point elevation & freezing-point depression

$$\Delta T_{\mathrm{b}} = K_{\mathrm{b}} m \tag{9}$$

$$\Delta T_{\mathrm{f}} = -K_{\mathrm{f}} m \tag{10}$$

2.3.3 Osmotic pressure

$$\pi = cRT \tag{11}$$

2.4 Volatile Solutes

2.4.1 Henry's Law

$$P_2 = k_2 X_2 \tag{12}$$

3 Chemical Equilibrium

3.1 Law of Mass Action

For the reaction $a\mathrm{A} + b\mathrm{B} \rightleftharpoons c\mathrm{C} + d\mathrm{D}$ at equilibrium

$$\frac{P_{\mathrm{C}}^c P_{\mathrm{D}}^d}{P_{\mathrm{A}}^a P_{\mathrm{B}}^b} = K \tag{13}$$

$$\frac{[\mathrm{C}]^c[\mathrm{D}]^d}{[\mathrm{A}]^a[\mathrm{B}]^b} = K \tag{14}$$

3.2 Reaction Quotient

For the reaction $a\mathrm{A} + b\mathrm{B} \rightleftharpoons c\mathrm{C} + d\mathrm{D}$

$$\frac{P_{\mathrm{C}}^c P_{\mathrm{D}}^d}{P_{\mathrm{A}}^a P_{\mathrm{B}}^b} = Q \tag{15}$$

$$\frac{[\mathrm{C}]^c[\mathrm{D}]^d}{[\mathrm{A}]^a[\mathrm{B}]^b} = Q \tag{16}$$

4 Acid-Base Equilibria

4.1 Ion product constant for water

$$[\mathrm{H_3O^+}][\mathrm{OH^-}] = K_{\mathrm{w}} \tag{17}$$

4.2 Definition of pH

$$\mathrm{pH} = -\log_{10}[\mathrm{H_3O^+}] \tag{18}$$

4.3 Ionization or dissociation constant

For a weak acid or a weak base

$$K_{\mathrm{a}} = \frac{[\mathrm{H_3O^+}][\mathrm{A^-}]}{[\mathrm{HA}]} \tag{19}$$

$$K_{\mathrm{b}} = \frac{[\mathrm{HB^+}][\mathrm{OH^-}]}{[\mathrm{B}]} \tag{20}$$

4.4 Buffer solution

$$\mathrm{pH} = \mathrm{p}K_\mathrm{a} - \log_{10} \frac{[\mathrm{HA}]_0}{[\mathrm{A}^-]_0} \tag{21}$$

5 Solubility and Precipitation Equilibria

5.1 Solubility product

For the reaction $\mathrm{M}_m\mathrm{X}_n(s) \rightleftharpoons m\mathrm{M}^{n+}(aq) + n\mathrm{X}^{m-}(aq)$ at equilibrium

$$K_\mathrm{sp} = [\mathrm{M}^{n+}]^m[\mathrm{X}^{m-}]^n \tag{22}$$

5.2 Formation of Complex Ions

For the reaction $m\mathrm{M}^{x+}(aq) + n\mathrm{L}(aq) \rightleftharpoons \mathrm{M}_m\mathrm{L}_n^{x+}(aq)$ at equilibrium

$$K_\mathrm{f} = \frac{[\mathrm{M}_m\mathrm{L}_n^{x+}]}{[\mathrm{M}^{x+}]^m[\mathrm{L}]^n} \tag{23}$$

6 Thermodynamic Processes and Thermochemistry

6.1 1st Law of Thermodynamics

$$\Delta U = q + w \tag{24}$$

6.1.1 P-V work

$$w = -P_\mathrm{ext}\Delta V \tag{25}$$

6.1.2 Heat

$$q = C\Delta T \tag{26}$$
$$q_\mathrm{V} = nc_\mathrm{V}\Delta T \tag{27}$$
$$q_\mathrm{P} = nc_\mathrm{P}\Delta T \tag{28}$$
$$q = mc_\mathrm{s}\Delta T \tag{29}$$

6.2 Enthalpy

$$H = U + PV \tag{30}$$
$$\Delta H = \Delta U + \Delta(PV) \quad \text{(in general)} \tag{31}$$
$$\Delta H = q_\mathrm{P} = \Delta U + P\Delta V \quad \text{(isobaric)} \tag{32}$$

6.3 Heat Capacity of Ideal Gases

$$c_{\mathrm{V}} = \frac{3}{2}R \qquad \text{monatomic ideal gas} \tag{33}$$

$$c_{\mathrm{P}} = c_{\mathrm{V}} + R \qquad \text{any ideal gas} \tag{34}$$

6.4 Standard Enthalpy Change of Reaction

$$\Delta H^\circ = \sum_{i=1}^{\text{product}} n_i \Delta H^\circ_{\mathrm{f},i} - \sum_{j=1}^{\text{reactant}} n_j \Delta H^\circ_{\mathrm{f},j} \tag{35}$$

6.5 Reversible Processes in Ideal Gases

6.5.1 Reversible Isothermal Processes

$$q = -w = nRT \ln \frac{V_2}{V_1} \tag{36}$$

$$\Delta T = \Delta U = \Delta H = 0 \tag{37}$$

6.5.2 Reversible Adiabatic Processes

$$q = 0 \tag{38}$$

$$\gamma = c_{\mathrm{P}}/c_{\mathrm{V}} \tag{39}$$

$$T_1 V_1^{\gamma-1} = T_2 V_2^{\gamma-1} \tag{40}$$

$$P_1 V_1^{\gamma} = P_2 V_2^{\gamma} \tag{41}$$

7 Spontaneous Processes and Thermodynamic Equilibrium

7.1 Boltzmann's Definition of Entropy

$$S = k_{\mathrm{B}} \ln \Omega \tag{42}$$

7.2 2nd Law of Thermodynamics

7.2.1 Thermodynamic Efficiency of the Carnot Cycle

$$\epsilon = \frac{w}{q_{\mathrm{h}}} = 1 - \frac{T_{\mathrm{l}}}{T_{\mathrm{h}}} \tag{43}$$

7.2.2 Clausius's Definition of Entropy

$$\Delta S = \int_{\mathrm{i}}^{\mathrm{f}} \frac{\mathrm{d}q_{\mathrm{rev}}}{T} \tag{44}$$

7.2.3 Isothermal Processes

$$\Delta S = \frac{q_{\text{rev}}}{T} = nR \ln \frac{V_2}{V_1} \qquad \text{(compression or expansion)} \tag{45}$$

$$\Delta S_{\text{PT}} = \frac{q_{\text{rev}}}{T_{\text{trans}}} = \frac{\Delta H_{\text{trans}}}{T_{\text{trans}}} \qquad \text{(phase transition)} \tag{46}$$

7.2.4 Adiabatic Processes

$$\Delta S = 0 \tag{47}$$

7.2.5 Isochoric Processes

$$\Delta S = \int_{T_1}^{T_2} \frac{\mathrm{d}q_{\text{rev}}}{T} = \int_{T_1}^{T_2} \frac{nc_{\text{V}}\,\mathrm{d}T}{T} = nc_{\text{V}} \ln \frac{T_2}{T_1} \tag{48}$$

7.2.6 Isobaric Processes

$$\Delta S = \int_{T_1}^{T_2} \frac{\mathrm{d}q_{\text{rev}}}{T} = \int_{T_1}^{T_2} \frac{nc_{\text{P}}\,\mathrm{d}T}{T} = nc_{\text{P}} \ln \frac{T_2}{T_1} \tag{49}$$

7.2.7 Clausius Inequality

$$\Delta S \geq \frac{q}{T} \tag{50}$$

7.2.8 Surroundings and Universe

$$\Delta S_{\text{surr}} = \frac{-\Delta H_{\text{sys}}}{T_{\text{surr}}} \tag{51}$$

$$\Delta S_{\text{tot}} = \Delta S_{\text{sys}} + \Delta S_{\text{surr}} \tag{52}$$

7.3 3nd Law of Thermodynamics

7.3.1 Standard Molar Entropy

$$S^\circ = \int_0^{298.15\ \text{K}} \frac{c_{\text{P}}}{T}\,\mathrm{d}T + \sum \Delta S_{\text{phase}} \tag{53}$$

7.4 Gibbs Free Energy

$$G \equiv H - TS \tag{54}$$

$$\Delta G = \Delta H - T\Delta S \qquad \text{constant } T\&P \tag{55}$$

7.5 Thermodynamic Description of the Equilibrium State

7.5.1 Thermodynamic Equilibrium Constant

$$\Delta G = \Delta G^\circ + RT \ln Q \tag{56}$$
$$-\Delta G^\circ = RT \ln K(T) \tag{57}$$

7.5.2 van't Hoff Equation

$$\ln \frac{K_2}{K_1} = -\frac{\Delta H^\circ}{R}(\frac{1}{T_2} - \frac{1}{T_1}) \tag{58}$$

8 Electrochemistry

8.1 Electrostatic Potential Energy and Electrical Work

$$w_{\text{elec}} = \Delta E_{\text{P}} = -Q\Delta E_{\text{cell}} \tag{59}$$

8.2 Faraday's Law

$$n = \frac{It}{F} \tag{60}$$

8.3 Gibbs Free Energy

$$w_{\text{elec,rev}} = \Delta G = -nF\Delta E_{\text{cell}} \tag{61}$$
$$\Delta G^\circ = -nF\Delta E^\circ_{\text{cell}} \tag{62}$$

8.4 Cell Potential

$$\Delta E_{\text{cell}} = E_{\text{cathode}} - E_{\text{anode}} \tag{63}$$
$$\Delta E^\circ_{\text{cell}} = E^\circ_{\text{cathode}} - E^\circ_{\text{anode}} \tag{64}$$

8.5 Nernst Equation

$$\Delta E_{\text{cell}} = \Delta E^\circ_{\text{cell}} - \frac{RT}{nF} \ln Q \tag{65}$$
$$\Delta E_{\text{cell}} = \Delta E^\circ_{\text{cell}} - \frac{0.0592\ \text{V}}{n} \log_{10} Q\ (\text{at } 25^\circ\text{C}) \tag{66}$$

8.6 Equilibrium Constants

$$\ln K = \frac{nF}{RT}\Delta E^\circ_{\text{cell}} \tag{67}$$
$$\log_{10} K = \frac{n}{0.0592\ \text{V}}\Delta E^\circ_{\text{cell}}\ (\text{at } 25^\circ\text{C}) \tag{68}$$

8.7 Energy Level *vs.* Potential

$$\epsilon = (-e)E \tag{69}$$

9 Chemical Kinetics

9.1 Rate Law

$$\text{rate} = -\frac{\mathrm{d}[\mathrm{R}]}{\mathrm{d}t} = k[\mathrm{R}]^n \tag{70}$$

9.2 Integrated Rate Law

9.2.1 Zero Order Reaction

$$\text{rate} = -\frac{\mathrm{d}[\mathrm{R}]}{\mathrm{d}t} = k \tag{71}$$

$$[\mathrm{R}]_t = [\mathrm{R}]_0 - kt \tag{72}$$

9.2.2 First Order Reaction

$$\text{rate} = -\frac{\mathrm{d}[\mathrm{R}]}{\mathrm{d}t} = k[\mathrm{R}] \tag{73}$$

$$\ln[\mathrm{R}]_t = \ln[\mathrm{R}]_0 - kt \tag{74}$$

$$[\mathrm{R}]_t = [\mathrm{R}]_0 e^{-kt} \tag{75}$$

$$t_{\frac{1}{2}} = \frac{\ln 2}{k} = \frac{0.6931}{k} \tag{76}$$

9.2.3 Second Order Reaction

$$\text{rate} = -\frac{\mathrm{d}[\mathrm{R}]}{\mathrm{d}t} = k[\mathrm{R}]^2 \tag{77}$$

$$\frac{1}{[\mathrm{R}]_t} = kt + \frac{1}{[\mathrm{R}]_0} \tag{78}$$

9.3 Principle of Detailed Balance

$$K = \prod_i K_i = \prod_i \frac{k_i}{k_{-i}} \tag{79}$$

9.4 Arrhenius Equation

$$k = Ae^{-\frac{E_a}{RT}} \tag{80}$$

$$\ln\frac{k_2}{k_1} = -\frac{E_a}{R}(\frac{1}{T_2} - \frac{1}{T_1}) \tag{81}$$

9.5 Michaelis-Menten Kinetics

$$K_{\mathrm{m}} = \frac{k_{-1} + k_2}{k_1} \tag{82}$$

$$\frac{\mathrm{d[P]}}{\mathrm{d}t} = k_2[\mathrm{ES}] = \frac{k_2[\mathrm{E_T}][\mathrm{S}]}{[\mathrm{S}] + K_{\mathrm{m}}} = \frac{V_{\max}[\mathrm{S}]}{[\mathrm{S}] + K_{\mathrm{m}}} \tag{83}$$

$$k_{\mathrm{cat}} \equiv k_2 = \frac{V_{\max}}{[\mathrm{E_T}]} \tag{84}$$

10 Structures and Bonding in Solids

10.1 Bragg Law

$$n\lambda = 2d\sin\theta \tag{85}$$

10.2 Unit Cell Volume

$$V_{\mathrm{c}} = abc\sqrt{1 - \cos^2\alpha - \cos^2\beta - \cos^2\gamma + 2\cos\alpha\cos\beta\cos\gamma} \tag{86}$$

10.3 Density of the Crystalline Element

$$\rho = \frac{n_{\mathrm{c}}M}{N_{\mathrm{A}}V_{\mathrm{c}}} \tag{87}$$

10.4 Born-Landé Equation

$$E = k\frac{Q_+Q_-}{r} \tag{88}$$

11 Polymeric Materials

11.1 Molecular Weight

$$w_i = iM_0 \tag{89}$$

$$M_{\mathrm{n}} = \frac{\sum_i n_i w_i}{\sum_i n_i} = \frac{\sum_i in_i}{\sum_i n_i}M_0 = \mathrm{DP_n}M_0 \tag{90}$$

$$M_{\mathrm{w}} = \frac{\sum_i n_i w_i^2}{\sum_i n_i w_i} = \frac{\sum_i i^2 n_i}{\sum_i in_i}M_0 \tag{91}$$

$$Đ = \frac{M_{\mathrm{w}}}{M_{\mathrm{n}}} \tag{92}$$